Digital Electronics (ECT-033) - Unit 3

Unit 3 - Sequential Circuits - notes in easy English, matched to the UTU syllabus.

Study Materials by Anas - study-materials-by-anas.vercel.app

What this unit covers: sequential circuits - circuits with memory: latches, flip-flops, registers, counters, and state-machine design.

Key ideas: Sequential outputs depend on inputs *and* the stored state. A **latch** is level-sensitive, a **flip-flop** edge-triggered. Know the four characteristic tables and equations: **SR** ($Q^+_1 = S + R'Q$), **JK** ($Q^+_1 = JQ' + K'Q$ - no invalid state), **D** ($Q^+_1 = D$), **T** ($Q^+_1 = T \oplus Q$). Converting one flip-flop to another uses excitation tables. **Registers** store words; shift registers move bits (SISO, SIPO, PISO, PIPO). **Counters**: asynchronous (ripple - simple but slow due to propagation delay) vs synchronous (all flip-flops share the clock - faster). Design steps for a synchronous counter: state diagram → state table → excitation table → K-maps → circuit. A mod-N counter counts N states.

Exam pointers: "design a mod-6 synchronous counter using JK flip-flops" style questions carry the big marks - practise the full design sequence once, slowly. Characteristic tables/equations of the four flip-flops and the ripple vs synchronous comparison are guaranteed theory.

Extra note - State Reduction & ASM Charts - simple notes:

Why reduce states? Two states are **equivalent** when, for every input, they give the same output and move to equivalent next-states. Merging them gives a smaller circuit (fewer flip-flops, less hardware) that behaves identically.

State reduction - steps (row-matching / implication table):

1. Write the full state table (present state, next state for each input, output).
2. Find pairs of states with identical outputs and identical (or equivalent) next-states for every input.
3. Merge each equivalent pair into one state and rewrite the table. Repeat until no pair merges.
4. Result: the **minimal state table** - same behaviour, fewest states.

> *Like finding two apps on your phone that do the exact same job - delete one, nothing changes for you, but the phone runs lighter.*

ASM chart (Algorithmic State Machine): a flowchart-style way to describe a sequential circuit. Three blocks: **state box** (one state, lists Moore outputs), **decision box** (diamond - checks an input condition, branches), **conditional output box** (oval - Mealy outputs that depend on state + input). Each path through state box -> decisions -> next state is one clock cycle. To design: draw the ASM chart, convert each state box to a state, read transitions and outputs straight into the state table.

Exam tip: for reduction, always show the equivalence pairs you merged; for ASM, label every box type - examiners deduct for missing state-box names.

Extra note - Pulse Train & PRBS Generator - made simple:

Pulse train generator = a circuit that outputs a **fixed pattern** on every clock tick, again and again. Example: 1, 0, 1, 1, 0 ... then back to 1, 0, 1, 1, 0 ... repeat.

> *Imagine people standing in a line. On every "tick", everyone steps forward and the person at the end of the line steps out. The pattern that comes out is the pulse train.*

How to build it: a shift register (a line of flip-flops) plus a feedback wire that decides what goes back into the start of the line.

Small example: want the pattern `101` (length 3). Take 3 flip-flops: Q1 Q2 Q3. On every tick all bits shift forward and the feedback puts the right new bit at the start. The pattern repeats forever: 101, 101, 101...

Clock generation: a T flip-flop **halves** the input frequency. This is how a slow, steady clock is made from a crystal oscillator.

PRBS (Pseudo-Random Binary Sequence): a magic box that gives "random-looking" bits - but actually from a fixed rule, so it is **pseudo (fake) random**.

How it is built: some middle outputs of the shift register are **XOR**ed and fed back. This is called an **LFSR (Linear Feedback Shift Register)**. With n flip-flops you can get a maximum-length sequence of `2^n - 1` bits before it repeats.

Where used: testing communication links, noise generation, bit-error-rate testing.

Exam tip: draw the shift register, mark the XOR feedback taps, and trace 4-5 clock cycles in a table - that is the standard full-marks answer.